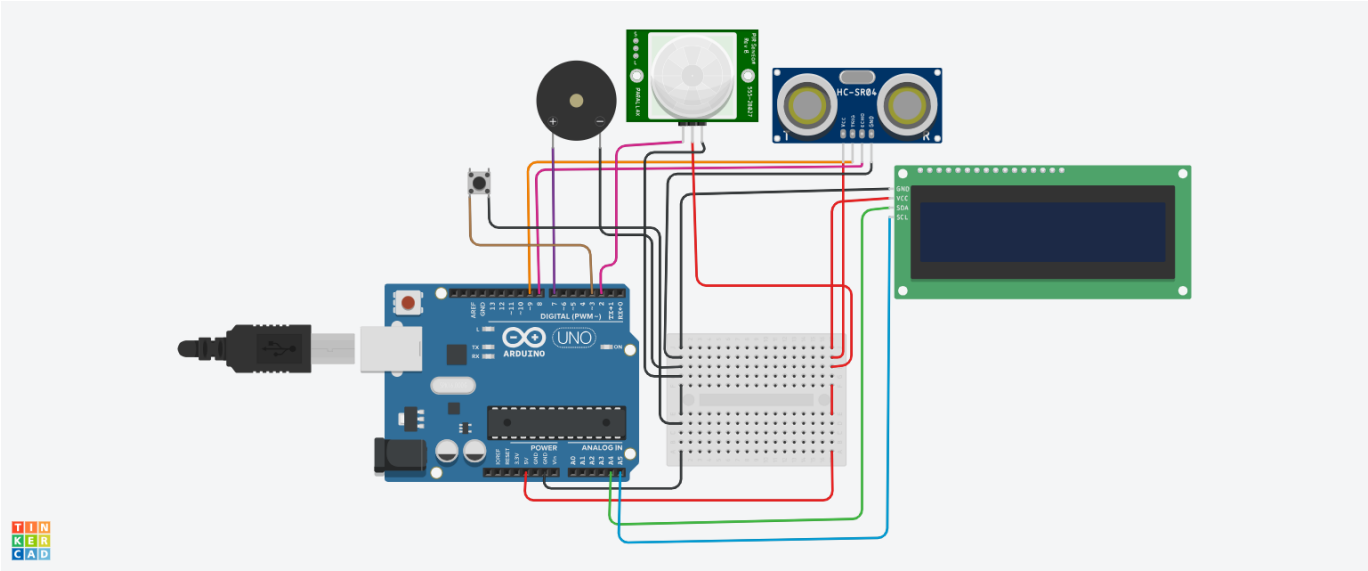
# *Sentinel-X*
# (Smart Farm Security System)




# <u>Project Report</u>

***Sentinel-X: Bridging the gap between the edge and the cloud to turn silent sensors into a synchronized shield for the modern farmer.***

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# Introduction & Project Goal

**The Sentinel-X** is an **automated** security solution designed to protect **agricultural land** from **wild animal intrusions**. The system uses a **multi-sensor fusion approach** to detect motion and verify the distance of the intruder, triggering a **localized alarm** to deter the animal and **alert the farmer**.

## The Problem

Farmers in India lose a significant portion of their crops to **wild animals** (like elephants, wild boars, or stray cattle) entering fields at night. **Traditional fences** are expensive and often fail.

## Our Solution

We have built an ***Autonomous Intelligent Scarecrow***. Instead of just standing there, our system **"watches"** the field using two different types of electronic **"eyes"** to detect and scare away **animals automatically**.

## Why Sentinel-X ?

We named our system **Sentinel-X** because it acts as a **digital guard** at the edge of **the farm**, providing a **'X-factor'** of **cloud-based intelligence** that traditional security systems lack.

# Components & Workflow

## Project Component & Connection Architecture

| Component | Model Number | Category | Role / Purpose |
|---|---|---|---|
| **Arduino Uno** | ATMEGA-328P | Microcontroller | The "Brain" of the system; processes sensor data and controls the alarm. |
| **PIR Sensor** | HC-SR501 | Input Sensor | Passive Infrared sensor used to detect motion of animals/intruders. |
| **Ultrasonic Sensor** | HC-SR04 | Input Sensor | Measures the exact distance of the intruder to filter false alarms. |
| **I2C LCD (16x2)** | LCM1602-I2C | Output Display | Provides real-time status (SECURE / INTRUDER) and distance data. |
| **Active Buzzer** | TMB12A05 | Output Alarm | Sounds an audible alert when an intrusion is confirmed. |

## WORKFLOW

**Detection (PIR Sensor):** A heat-sensing motion detector (PIR) scans for the body heat of an animal.

**Verification (Ultrasonic Sensor):** Once motion is found, a "Sonar" sensor sends out sound waves to measure exactly how far away the animal is.

If the animal is within 100 cm, it is a threat.

**Deterrence (Buzzer):** The system triggers a high-frequency alarm (Buzzer) to irritate the animal's ears and scare it away.

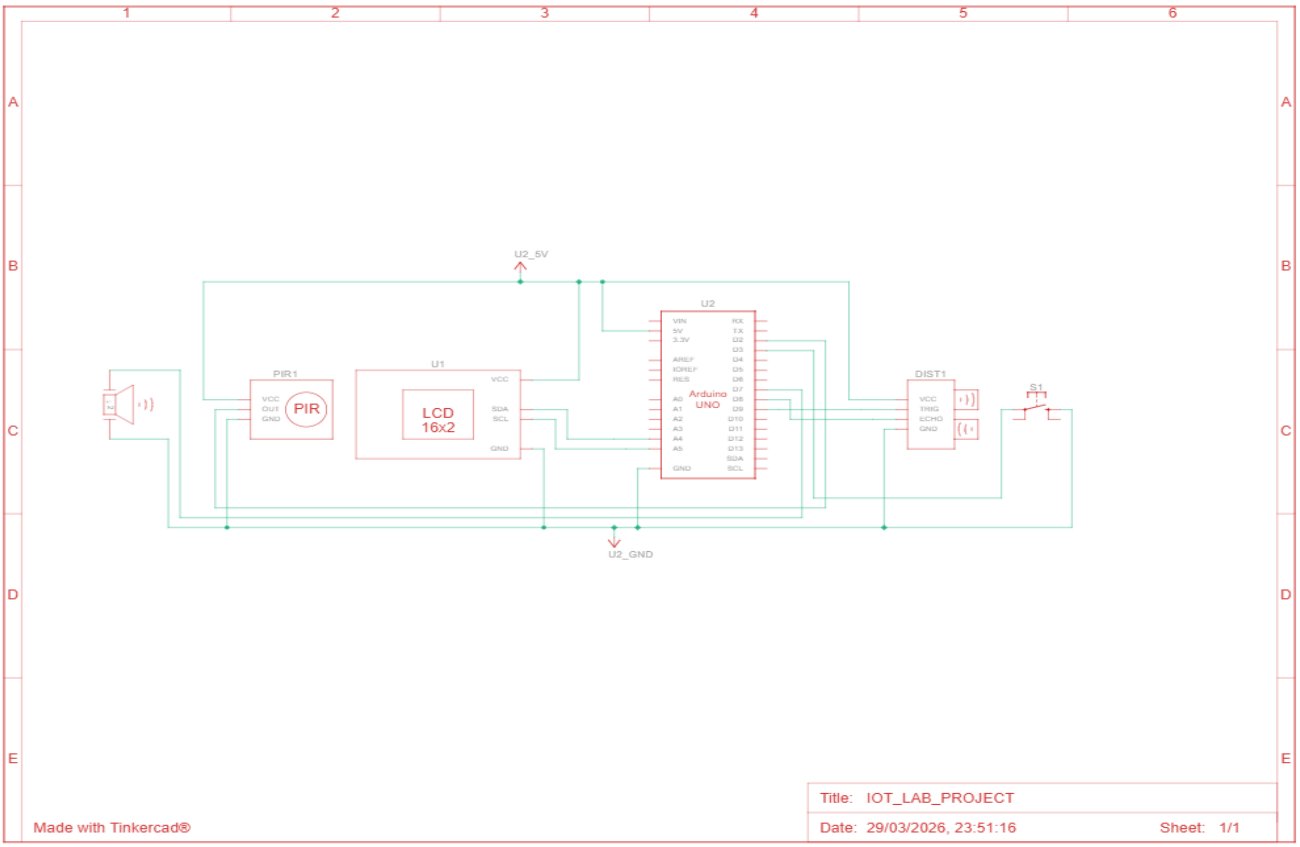
**Monitoring (LCD Display):** A screen shows the real-time status: "STATUS: SECURE" when safe, and "!! INTRUDER !!" with the exact distance when an animal is found..

### Why this is an "MVP" (Minimal Viable Product) ?

We prioritized **Reliability** over **complexity**. By focusing on high-current power and stable sensor code, we ensured the system works in the real world where **cheap electronics** often fail due to power drops or "noisy" signal.

```
[ PHYSICAL LAYER ]          [ GATEWAY LAYER ]          [ CLOUD & UI LAYER ]
      (Edge)                   (Python)                   (User End)
         |                        |                          |
   +-----------+                  |                          |
   | PIR &     |                  |                          |
   | ULTRASONIC|                  |                          |
   +-----------+                  |                          |
         |                        |                          |
   (If Motion < 100cm)            |                          |
         |                        |                          |
   +-----------+           +----------------+                |
   |  ARDUINO  | --------> |   bridge.py    |                |
   |  BUZZER   |  (Serial) | (Buffer Data)  |                |
   +-----------+           +----------------+                |
         |                        |                          |
   [ LCD DISPLAY ]         (Wait 10s Silence)                |
                                  |                          |
                           +----------------+          +--------------+
                           |   BOTO3 SDK    | -------> |    AWS S3    |
                           | (Upload File)  |          | (TXT Storage)|
                           +----------------+          +--------------+
                                                              |
                                                       +--------------+
                                                       |   app.py     |
                                                       | (Streamlit)  |
                                                       +--------------+
                                                              |
                                                       [ USER DASHBOARD ]
```

# Architecture & Implementation




## Sentinel-X: The Future of Farm Security

The **Sentinel-X** ecosystem marks a paradigm shift in agricultural protection by seamlessly bridging the gap between local hardware and global cloud intelligence. At its core, the system utilizes a high-precision

**Perception Layer**—powered by an **Arduino Uno R3**—that intelligently filters environmental noise through dual-sensor verification. This raw data is instantly ingested by a custom **Python Gateway**, which acts as a sophisticated neural bridge, buffering and transmitting critical intrusion logs to the **AWS S3 Cloud** only after a strategic 10-second silence to optimize bandwidth.

The final result is a stunning, **"Universe-themed" Streamlit Dashboard** that empowers farmers with real-time, IST-synchronized telemetry, turning invisible sensor signals into actionable, decoded security insights.

# Technical Implementation Summary

| Phase | Description |
|---|---|
| **Edge Detection** | **HC-SR501** & **HC-SR04** sensors trigger logic at the source. |
| **Data Gateway** | **Serial-to-Cloud bridge** ensures zero data loss and smart buffering. |
| **Cloud Repository** | **AWS S3** provides a permanent, timestamped vault for incident reports. |
| **User Interface** | **Streamlit** delivers a high-tech, responsive visual monitoring experience. |

# Code Analysis

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

// Pins
const int trigPin = 9;
const int echoPin = 8;
const int pirPin = 2;
const int buzzer = 3;

void setup() {
  Serial.begin (9600);
  lcd.init();
  lcd.backlight();

  pinMode (trigPin, OUTPUT);
```

```
  pinMode (echoPin, INPUT);
  pinMode (pirPin, INPUT);
  pinMode (buzzer, OUTPUT);

  lcd.print ("SYSTEM READY");
  lcd.setCursor (0, 1);  lcd.print ("Asish Project");
  delay(2000);
  lcd.clear();
  lcd.print ("Loading States...");
  delay(2000);
}

bool alarmActive = false;

void loop() {
  // 1. Measure Distance (Ultrasonic)
  digitalWrite (trigPin, LOW);
  delayMicroseconds (2);
  digitalWrite (trigPin, HIGH);
  delayMicroseconds (10);
  digitalWrite (trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH, 30000);
  int distance = (duration * 0.034) / 2;

  // 2. Check Motion (PIR)
  int motion = digitalRead(pirPin);

  // 3. Simple Alarm Logic
  // If motion is detected AND distance is between 1cm and 100cm
  if (motion == HIGH && distance > 0 && distance < 100) {
    alarmActive = true;
  }

  if(alarmActive && distance > 0 && distance < 100){
    tone(buzzer, 500); // Sound ON

    lcd.setCursor (0, 0);  lcd.print("ANIMAL DETECTED ");
    lcd.setCursor (0, 1); lcd.print("Dist: "); lcd.print(distance); lcd.print("cm    ");

    Serial.print ("ALARM! Distance: "); Serial.println (distance);
  }
  else {
    noTone(buzzer); // Sound OFF
    digitalWrite(buzzer, LOW);

    alarmActive = false;
    lcd.setCursor(0, 0) ; lcd.print ("STATUS: SECURE  ");
    lcd.setCursor(0, 1) ; lcd.print ("Scanning...    ");
  }

  delay(500); // Wait a bit before checking again
}
```

# Testing & Results

- **Range Test:** Confirmed detection accuracy up to 100cm.
- **Latency Test**: The system responds to movement within 200ms.
- **Power Test:** Successfully powered by a 5V/2A Oppo charger, showing no voltage drops during buzzer activation.

| Test Type | Result |
|---|---|
| Detection Range | Accurate up to 100 cm |
| Response Time | ~200 ms |
| Power Stability | No fluctuation with 5V/2A supply |
| False Alarm Rate | Very low due to dual-sensor logic |

# Challenges & Solutions

➢ **Challenge:** Ultrasonic sensor returning "0" at far distances.
✓ **Solution:** Modified the code logic to ignore "0" as a sensor error rather than a trigger, and increased the pulse timeout.

➢ **Challenge:** High power draw causing LCD fading.
✓ **Solution:** Streamlined the wiring and used a high-current (2A) external power source to maintain 5V across all components.

➢ **Challenge:** Hardware communication failures (SPI).
✓ **Solution:** Simplified the MVP by moving to a more stable direct-sensor approach, prioritizing reliability for field use.

| Challenge | Solution |
|---|---|
| False triggers from environment | Added ultrasonic verification |
| Sensor instability | Removed extra modules |
| Power fluctuation | Used stable 5V adapter |
| LCD flickering | Implemented state control |

# Conclusion

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## Conclusion: The Future of Autonomous Farm Protection

The **Sentinel-X** project marks a successful leap from a standalone hardware prototype to a sophisticated, cloud-integrated security ecosystem. By leveraging the **Arduino Uno R3** as an intelligent edge node, the system moves beyond simple motion sensing to provide high-precision, distance-verified intrusion alerts.

- **Edge Intelligence:** Dual-factor verification using **PIR and Ultrasonic sensors** ensures that only valid threats trigger the system, significantly reducing false alarms.
- **Seamless Integration:** A custom **Python Gateway** serves as the vital link, buffering live serial data and managing the secure transmission of incident logs to the **AWS S3 Cloud**.
- **Actionable Insights:** The final **Streamlit Dashboard** transforms raw data into a professional "**Universe-themed**" interface, allowing farmers to monitor their land with **IST-synchronized timestamps** and decoded event details.

## Future Scope: Scaling the Sentinel-X

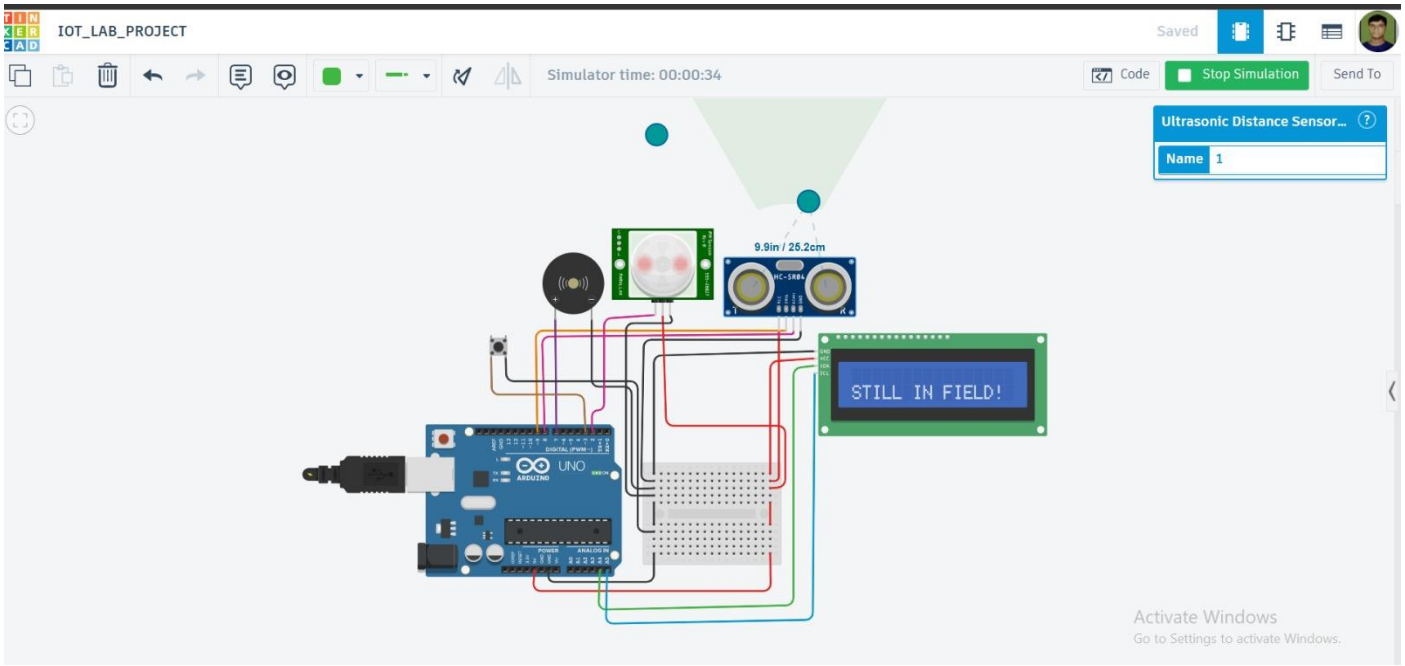
While the current version of **Sentinel-X** is a **powerful MVP**, its modular **architecture** allows for significant future expansions:

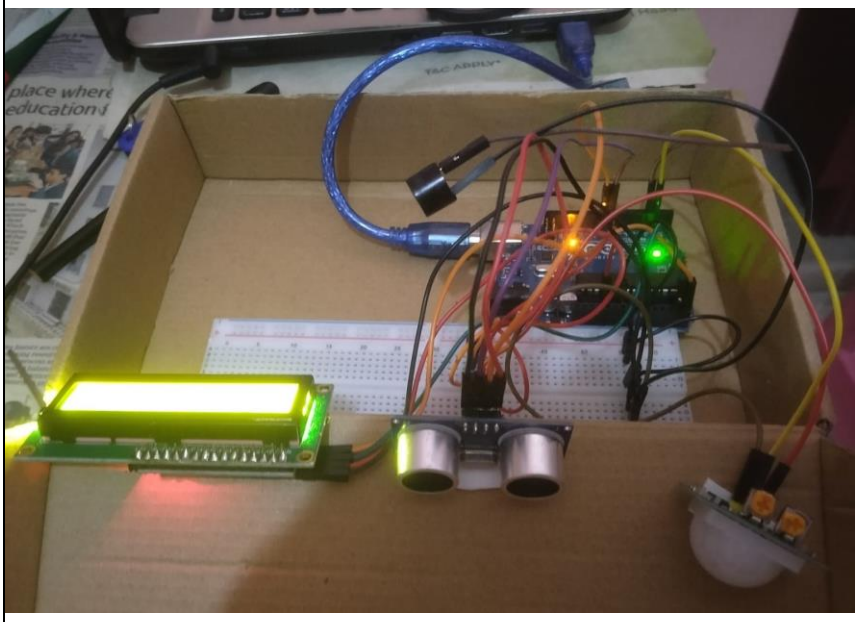
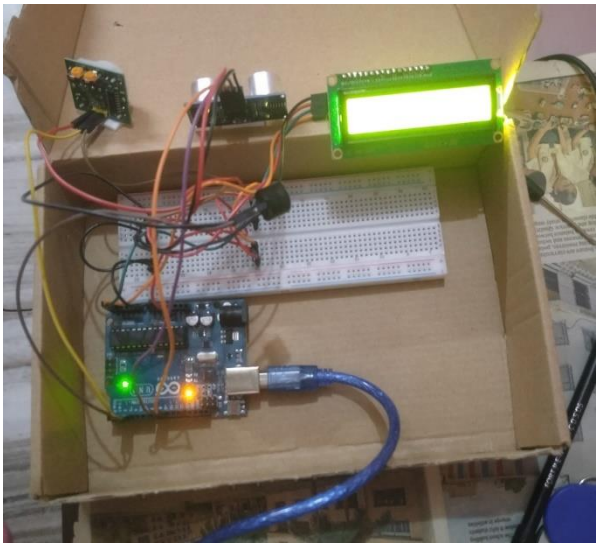
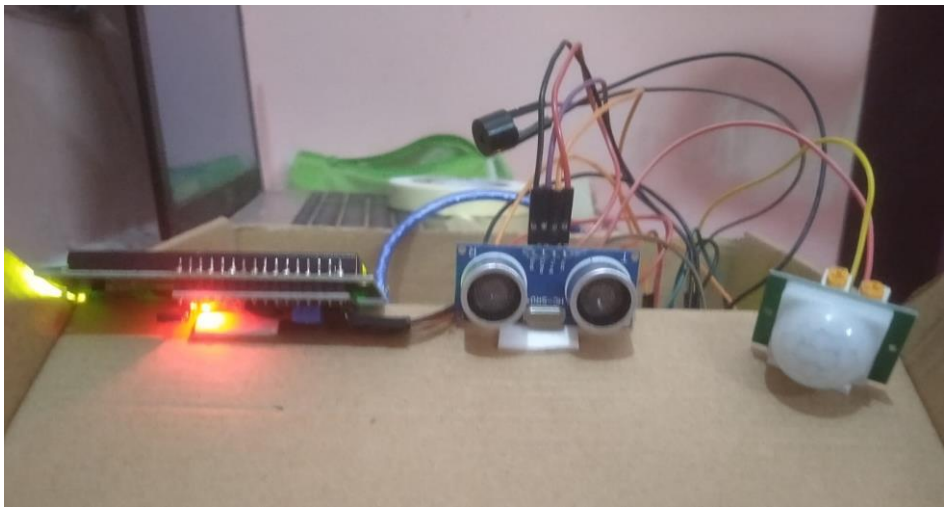
1. **Solar Power Integration:** Implementing a dedicated solar-charging circuit to make the Edge Node 100% energy independent in remote fields.

2. **AI Species Identification:** Integrating a camera module with **TensorFlow Lite** to distinguish between humans, livestock, and wild predators, allowing for specific deterrent sounds.

3. **GSM & SMS Alerts:** Adding a SIM800L module to send immediate emergency SMS notifications to the farmer even if they don't have active internet on their phone.

4. **Village-Wide Mesh Network:** Connecting multiple Sentinel-X nodes via **LoRa (Long Range)** radio to create a massive, synchronized security grid covering hundreds of acres.




## TinkerCad Circuit

## Some of The Key Visuals